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D20A-40

Experiment :05
Blockchain Lab

Aim: Hands on Solidity Programming Assignments for creating Smart Contracts
Theory:

1. Relevance of require Statements in Solidity Programs

In Solidity, the require statement acts as a **guard condition** within functions. It ensures that only valid inputs or authorized users can execute certain parts of the code. If the condition inside require is not satisfied, the function execution stops immediately, and all state changes made during that transaction are reverted to their original state. This rollback mechanism ensures that invalid transactions do not corrupt the blockchain data.

For example, in a **Voting Smart Contract**, require can be used to check:

- Whether the person calling the function has the right to vote (require(voters[msg.sender].weight > 0, "Has no right to vote");).
- Whether a voter has already voted before allowing them to vote again.
- Whether the function caller is the **chairperson** before granting voting rights.

Thus, require statements enforce **security, correctness, and reliability** in smart contracts. They also allow developers to attach error messages, making debugging and contract interaction easier for users.

2. Keywords: mapping, storage, and memory

- **mapping**:
  A mapping is a special data structure in Solidity that links keys to values, similar to a hash table. Its syntax is mapping(keyType => valueType). For example: mapping(address => Voter) public voters;

  Here, each address (Ethereum account) is mapped to a Voter structure. Mappings are very useful for contracts like **Ballot**, where you need to associate voters with their data (whether they voted, which proposal they chose, etc.). Unlike arrays, mappings do not have a length property and cannot be iterated over directly, making them **gas efficient** for lookups but limited for enumeration.

- **storage**:
  In Solidity, storage refers to the **permanent memory** of the contract, stored on the Ethereum blockchain. Variables declared at the contract level are stored in storage by default. Data stored in storage is persistent across

transactions, which means once written, it remains available unless explicitly modified. However, because writing to blockchain storage consumes gas, it is more expensive. For example, a voter's information saved in the voters mapping remains available throughout the contract's lifecycle.

- **memory**:
  In contrast, memory is **temporary storage**, used only for the lifetime of a function call. When the function execution ends, the data stored in memory is discarded. Memory is mainly used for temporary variables, function arguments, or computations that don't need to be permanently stored on the blockchain. It is cheaper than storage in terms of gas cost. For instance, when handling proposal names or temporary string manipulations, memory is often used.

Thus, a smart contract developer must **balance between storage and memory** to ensure efficiency and cost-effectiveness.

3. Why bytes32 Instead of string?

In earlier implementations of the Ballot contract, bytes32 was used for proposal names instead of string. The reason lies in **efficiency and gas optimization**.

- **bytes32** is a **fixed-size type**, meaning it always stores exactly 32 bytes of data. This makes storage simple, comparison operations faster, and gas costs lower. However, it limits proposal names to 32 characters, which is not very flexible for user-friendly names.
- **string** is a **dynamically sized type**, meaning it can store text of variable length. While it is easier for developers and users (since names can be written normally), it requires more complex handling inside the Ethereum Virtual Machine (EVM). This increases gas usage and may slow down comparisons or manipulations.

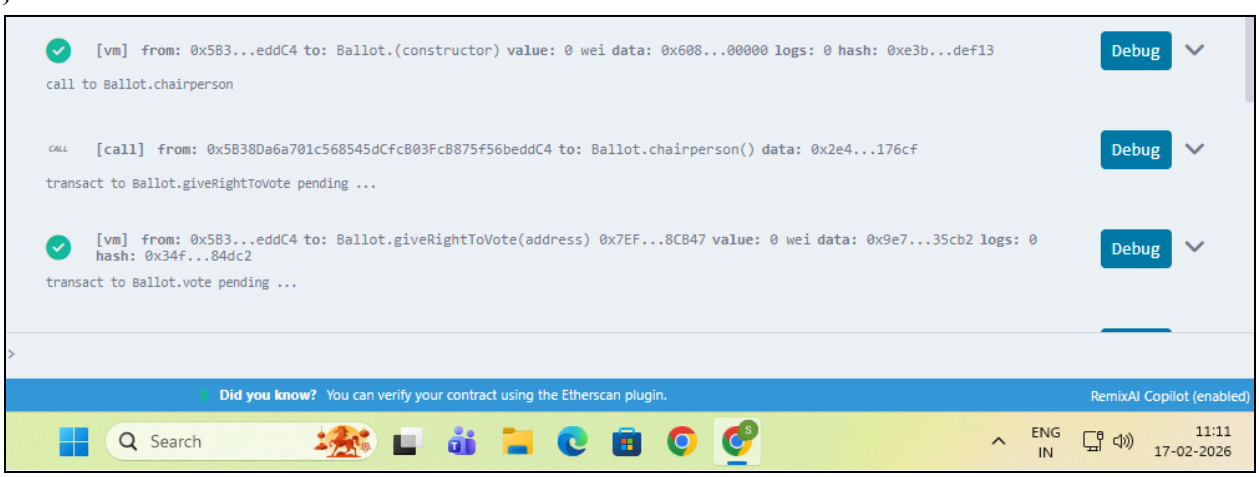
To make the system more user-friendly, modern implementations of the Ballot contract often convert from bytes32 to string. Tools like the **Web3 Type Converter** help developers easily switch between these two types for deployment and testing.

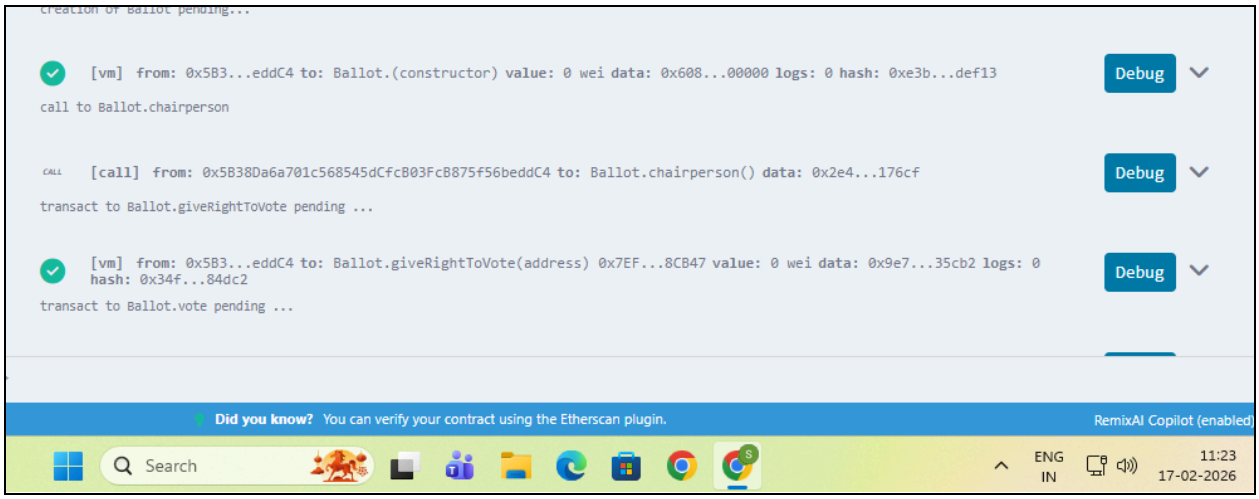
In summary, bytes32 is used when performance and gas efficiency are priorities, while string is preferred for readability and ease of use.

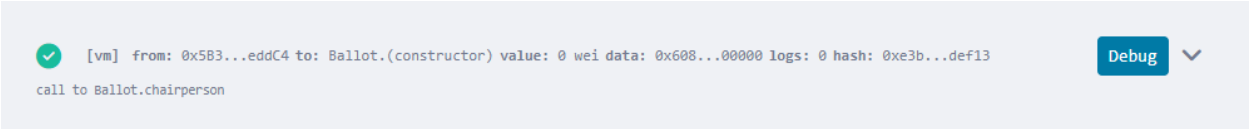
CODE:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract Ballot {

    struct Voter {
        uint weight;
        bool voted;
        address delegate;
        uint vote;
    }

    struct Proposal {
        string name;      // changed from bytes32 to string
        uint voteCount;
    }

    address public chairperson;
    mapping(address => Voter) public voters;
    Proposal[] public proposals;
    constructor(string[] memory proposalNames) {
        chairperson = msg.sender;
        voters[chairperson].weight = 1;

        for (uint i = 0; i < proposalNames.length; i++) {
            proposals.push(Proposal({
                name: proposalNames[i],
                voteCount: 0
            }));
        }
    }
    function giveRightToVote(address voter) public {
        require(msg.sender == chairperson, "Only chairperson can give right.");
        require(!voters[voter].voted, "The voter already voted.");
        require(voters[voter].weight == 0);
        voters[voter].weight = 1;
    }
    function vote(uint proposal) public {
```

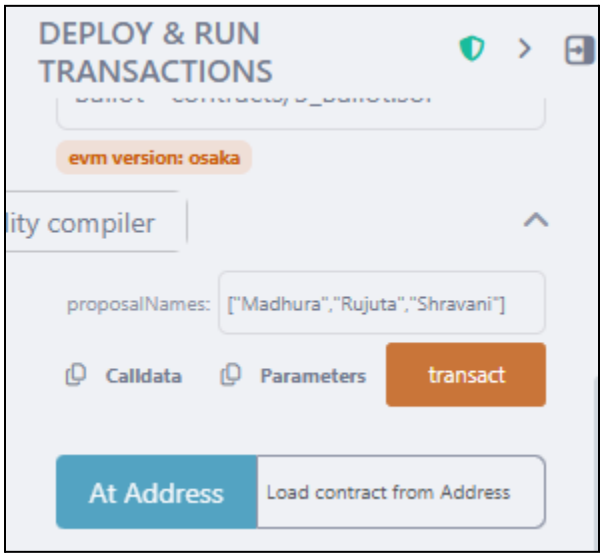
```
        Voter storage sender = voters[msg.sender];
        require(sender.weight != 0, "Has no right to vote");
        require(!sender.voted, "Already voted.");

        sender.voted = true;
        sender.vote = proposal;
        proposals[proposal].voteCount += sender.weight;
    }
    function winningProposal() public view returns (uint winningProposal_) {
        uint winningVoteCount = 0;

        for (uint p = 0; p < proposals.length; p++) {
            if (proposals[p].voteCount > winningVoteCount) {
                winningVoteCount = proposals[p].voteCount;
                winningProposal_ = p;
            }
        }
    }
    function winnerName() public view returns (string memory winnerName_) {
        winnerName_ = proposals[winningProposal()].name;
    }
}
```





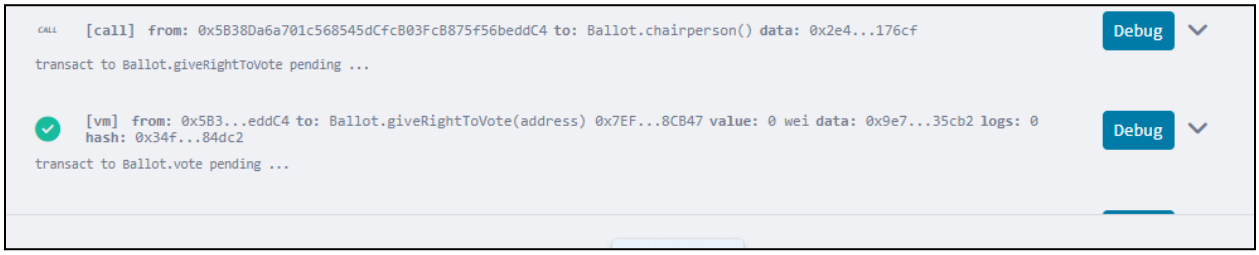


Transaction deployment and Proposal names:

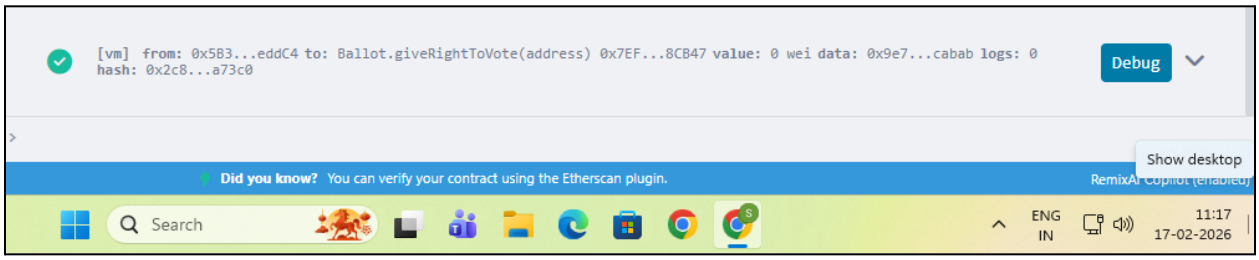


Right to Vote:

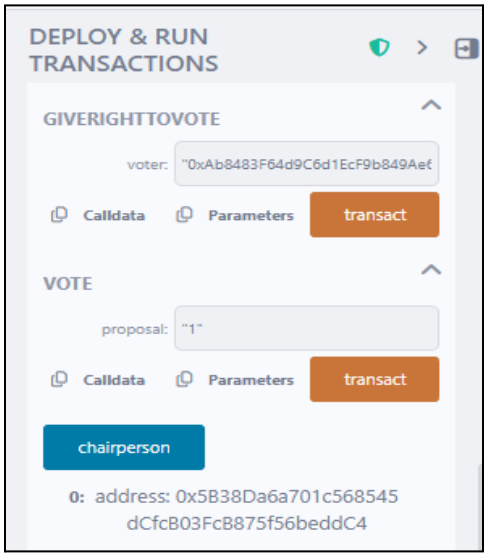
Voter 1:



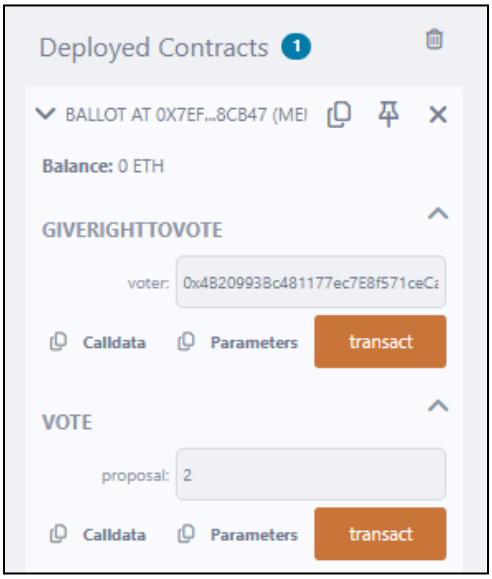
Voter 3:



VOTER 1:



VOTER 2 :



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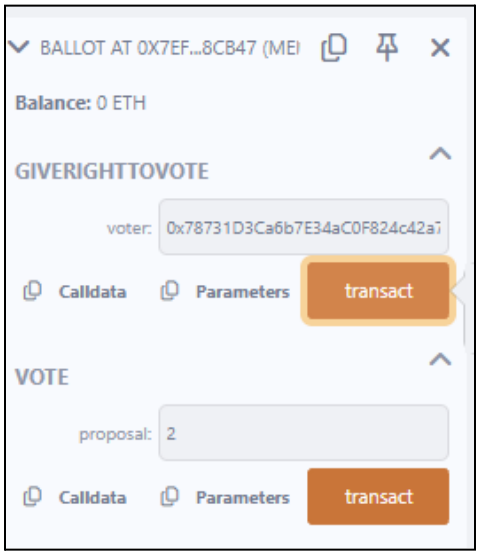
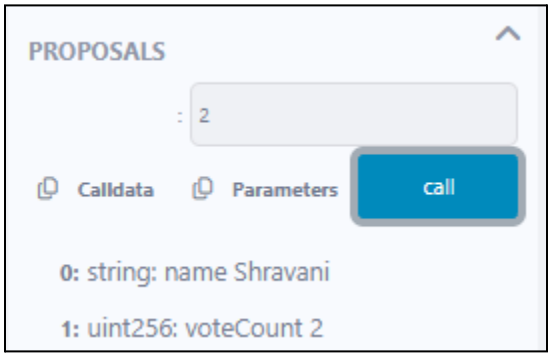
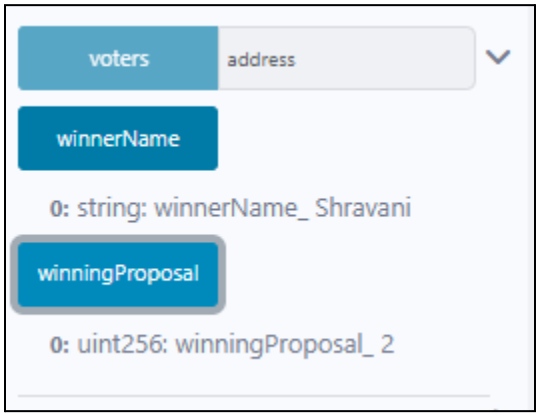
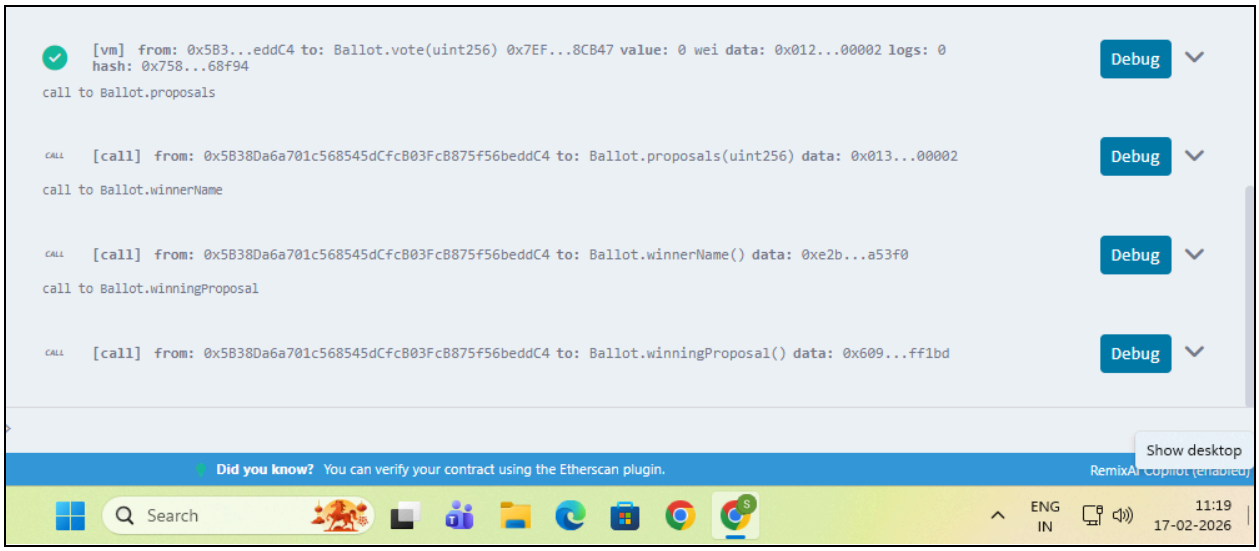


Checking Vote counts for proposal 2:



Winner name :





Conclusion

In this experiment, we successfully designed and deployed a **Voting/Ballot Smart Contract** using Solidity in the Remix IDE. The contract was modified by changing the proposal name data type from bytes32 to string, making it more readable and user-friendly.

We compiled and deployed the contract, assigned voting rights, cast votes from different accounts, and retrieved the winning proposal using smart contract functions. This experiment helped in understanding key blockchain concepts such as smart contracts, mappings, structs, arrays, constructors, and decentralized execution on the Ethereum Virtual Machine (EVM).

Thus, the objective of exploring blockchain concepts and designing a smart contract using Solidity (LO2) was successfully achieved.